

Special Relativity I

Lecture Notes

Galilean relativity, Minkowski spacetime, Lorentz transformations, four vectors, and relativistic fields

Metric signature $(+, -, -, -)$

Contents

1	Orientation	1
2	Newtonian spacetime and Galilean transformations	1
2.1	Events, observers, and inertial frames	1
2.2	Time translations	2
2.3	Spatial translations	2
2.4	Rotations	3
2.5	Galilean boosts	3
2.6	The combined transformation	4
3	The Galilei group	5
3.1	Groups	5
3.2	Composition law	5
3.3	Why this is a Lie group	7
4	Why Galilean invariance fails for light	8
5	The postulates of special relativity	9
6	Points, vectors, and coordinates	10
6.1	The affine plane	10
6.2	Bases and coordinate systems	10
7	Products of vectors	11
7.1	Inner and tensor products	11
7.2	The geometric product	12
8	What is a tensor?	14
9	Contraction and Einstein notation	15
9.1	Composition as contraction	15
9.2	Vectors and covectors	16
10	Four-vector notation and units	16
11	Invariants and Minkowski geometry	17
11.1	Euclidean invariants	17
11.2	The Minkowski metric	18

12 Events, intervals, and causal type	19
12.1 Coordinate dependence and invariant separation	19
12.2 Classification	19
12.3 Intervals and clock readings	21
13 Four-momentum and invariant mass	21
14 Lorentz and Poincaré transformations	22
15 The standard boost	24
16 Determining the boost matrix	24
17 Rapidity and relative velocity	26
18 Four-vectors and covectors	27
19 The metric and index operations	28
20 Lorentz transformations of fields	30
20.1 One event, two coordinate descriptions	30
20.2 Passive and active viewpoints	31

1 Orientation

SOURCE SLIDE 1

Special relativity is easiest to motivate by first understanding what fails in Newtonian kinematics. The chapter therefore begins with Galilean transformations and the structure of inertial frames. The conflict between Galilean velocity addition and the invariant speed of light then forces a new spacetime geometry.

To express that geometry precisely, the chapter next separates points from vectors and introduces covectors, tensors, and index notation. These tools lead naturally to Minkowski invariants, Lorentz transformations, rapidity, and the transformation laws of fields. Each part is meant to be read as a continuation of the preceding one.

Throughout the chapter, vector spaces are finite-dimensional and real unless stated otherwise. Explanations and statements are provided in full, while selected proofs and calculations are left in the open student-work spaces.

2 Newtonian spacetime and Galilean transformations

SOURCE SLIDE 2

2.1 Events, observers, and inertial frames

For the flat theories considered in this chapter, spacetime is modeled by a four-dimensional affine space $\mathcal{M}$ whose translation vector space is denoted by $\mathcal{V}$. Spacetime has no preferred origin. Its points and the coordinates used to label them must therefore be kept conceptually distinct. The affine-space language is developed more fully in Definition 6.1.

Definition 2.1: Event

An *event* is a point $X \in \mathcal{M}$. An inertial frame O assigns coordinates to events through an affine bijection

$$\chi_O : \mathcal{M} \longrightarrow \mathbb{R} \times \mathbb{R}^3, \quad X \longmapsto \chi_O(X) = (t, \mathbf{x}).$$

The tuple $(t, \mathbf{x})$ is a label for X , not the event itself. Another frame O' generally assigns the same event a different tuple $\chi_{O'}(X) = (t', \mathbf{x}')$.

An idealized physical occurrence localized to one place and one time is modeled by an event. A particle's history is a curve of events called its *worldline*.

Definition 2.2: Inertial reference frame

An *inertial reference frame* is a global affine spacetime coordinate system in which every free particle has constant coordinate velocity. If $\mathbf{x}(t)$ denotes such a particle's trajectory, this condition reads

$$\ddot{\mathbf{x}}(t) = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{x}(t) = \mathbf{x}_0 + \mathbf{u} t,$$

where $\mathbf{u} \in \mathbb{R}^3$ is constant.

The inertial coordinates used here have nonrotating Cartesian spatial axes and synchronized clocks at rest in the frame. Different frames use the same units of length and time. These conventions exclude arbitrary rescalings or nonstandard clock synchronizations from the transformations considered

below. An observer is represented by a worldline in spacetime. An observer at rest in a given frame follows a worldline of constant spatial coordinates. A frame supplies coordinates beyond that one worldline.

Galilean spacetime supplements the affine structure with an absolute notion of elapsed time and a Euclidean geometry on each surface of simultaneity. Let O and O' be inertial frames assigning coordinates $(t, \mathbf{x})$ and $(t', \mathbf{x}')$ to the same event. Their coordinates may differ through four elementary operations.

2.2 Time translations

The first freedom is the choice of zero on the common absolute time axis. The corresponding coordinate change is

$$t' = t + \tau, \quad \mathbf{x}' = \mathbf{x}, \quad \tau \in \mathbb{R}.$$

Although the time coordinate changes, an elapsed time does not.

Proposition 2.3: Time intervals are invariant

For two events with time coordinates t_1, t_2 , every Galilean transformation satisfies

$$\Delta t' = t'_2 - t'_1 = \Delta t.$$

In particular, simultaneity, $\Delta t = 0$, is observer independent in Galilean spacetime.

Student work – proof

Prove Proposition 2.3.

2.3 Spatial translations

The second freedom is the choice of spatial origin. A translation by the constant vector $\mathbf{a}$ is written

$$t' = t, \quad \mathbf{x}' = \mathbf{x} + \mathbf{a}, \quad \mathbf{a} \in \mathbb{R}^3.$$

Coordinates of individual points change, but relative displacement does not.

Proposition 2.4: Translations preserve relative displacement

If $\Delta \mathbf{x} = \mathbf{x}_2 - \mathbf{x}_1$, then

$$\Delta \mathbf{x}' = \Delta \mathbf{x}.$$

Student work – proof

Prove Proposition 2.4.

2.4 Rotations

The spatial axes may also differ by a proper rotation. Their coordinates are then related by

$$t' = t, \quad \mathbf{x}' = R\mathbf{x}, \quad R \in SO(3),$$

where

$$R^T R = \mathbf{I}, \quad \det R = 1.$$

The displacement between two simultaneous events consequently satisfies $\Delta \mathbf{x}' = R\Delta \mathbf{x}$.

Remark

The equation $R^T R = \mathbf{I}$ defines $O(3)$, whose elements include both proper rotations and reflections. Adding $\det R = 1$ selects $SO(3)$. These are precisely the orientation-preserving orthogonal transformations and form the component of $O(3)$ connected to the identity.

Proposition 2.5: Rotations preserve Euclidean separation

A proper rotation preserves the squared distance between any two spatial points. Thus

$$\|\Delta \mathbf{x}'\|^2 = \|\Delta \mathbf{x}\|^2.$$

Student work – proof

Prove Proposition 2.5.

2.5 Galilean boosts

Finally, inertial frames may move uniformly relative to one another. Following the convention used in the slides, a pure Galilean boost with parameter $\mathbf{v}$ is written

$$t' = t, \quad \mathbf{x}' = \mathbf{x} + \mathbf{v} t.$$

The symbol $\mathbf{v}$ is the *boost parameter*. It is important not to confuse this parameter with the velocity of the primed origin. Indeed, $\mathbf{x}' = \mathbf{0}$ implies $\mathbf{x} = -\mathbf{v}t$. Thus, if $\mathbf{V}$ is the physical velocity of O' relative to O , the convention above gives $\mathbf{V} = -\mathbf{v}$.

Proposition 2.6: Velocity and acceleration under a Galilean boost

For a particle with velocity $\mathbf{u} = d\mathbf{x}/dt$ and acceleration $\mathbf{a}_p = d\mathbf{u}/dt$,

$$\mathbf{u}' = \mathbf{u} + \mathbf{v}, \quad \mathbf{a}'_p = \mathbf{a}_p.$$

Consequently, the property of being an inertial frame is preserved by a Galilean boost.

Student work – derivation

Derive both transformation laws in Proposition 2.6.

2.6 The combined transformation

The four operations combine into a single affine transformation. Writing $g = (\tau, \mathbf{a}, \mathbf{v}, R)$, its action is

$$t' = t + \tau, \quad \mathbf{x}' = R\mathbf{x} + \mathbf{v}t + \mathbf{a}, \quad R \in SO(3).$$

It is affine rather than linear because the translation parameters need not vanish. When two event coordinates are subtracted, those constant terms cancel and the induced action on the displacement is linear. In particular,

$$\Delta t' = \Delta t, \quad \Delta \mathbf{x}' = R\Delta \mathbf{x} + \mathbf{v} \Delta t.$$

Hence spatial distances between *simultaneous* events are preserved by every Galilean transformation. For events at different times, their spatial separation generally changes under a boost.

The choice $R \in SO(3)$ restricts this family to transformations preserving spatial orientation and the direction of time. Spatial reflections could be included in a larger group, but are not included in the notation $\text{Gal}(3)$ used here.

3 The Galilei group

SOURCE SLIDE 3

3.1 Groups

Transformations between inertial frames can be composed and inverted, and the result is again a transformation between inertial frames. This is precisely the situation formalized by a group.

Definition 3.1: Group

A *group* is a pair $(G, \circ)$ consisting of a set G and a binary operation $\circ : G \times G \rightarrow G$. The codomain expresses closure. The operation is associative,

$$(g_3 \circ g_2) \circ g_1 = g_3 \circ (g_2 \circ g_1) \quad \text{for all } g_1, g_2, g_3 \in G.$$

There exists an identity $e \in G$ such that

$$e \circ g = g \circ e = g \quad \text{for every } g \in G.$$

For every $g \in G$, there exists an inverse $g^{-1} \in G$ satisfying

$$g^{-1} \circ g = g \circ g^{-1} = e.$$

A group is *abelian* if $g_2 \circ g_1 = g_1 \circ g_2$ for every pair of elements.

Example 3.2: Addition modulo three

The set $\{0, 1, 2\}$, with addition followed by taking the remainder modulo three, is an abelian group. Its identity is 0, and the inverse of 1 is 2.

Remark

Closure says that composing two allowed transformations produces another allowed transformation. It says nothing about whether two transformations commute. In a non-abelian group, both orders are valid but generally represent different transformations.

3.2 Composition law

Let

$$g_i = (\tau_i, \mathbf{a}_i, \mathbf{v}_i, R_i), \quad i = 1, 2,$$

act by the combined transformation above. The notation $g_2 \circ g_1$ means that g_1 acts first and g_2 acts second.

Proposition 3.3: Composition of Galilean transformations

The composite transformation is

$$g_2 \circ g_1 = (\tau_1 + \tau_2, R_2 \mathbf{a}_1 + \mathbf{v}_2 \tau_1 + \mathbf{a}_2, R_2 \mathbf{v}_1 + \mathbf{v}_2, R_2 R_1).$$

Student work – derivation

Derive Proposition 3.3.

Proposition 3.4: Identity and inverse

The identity is

$$e = (0, \mathbf{0}, \mathbf{0}, \mathbf{I}).$$

For $g = (\tau, \mathbf{a}, \mathbf{v}, R)$, the inverse is

$$g^{-1} = \left(-\tau, R^{-1}(\mathbf{v}\tau - \mathbf{a}), -R^{-1}\mathbf{v}, R^{-1} \right).$$

Student work – derivation

Derive Proposition 3.4.

Proposition 3.5: Associativity and noncommutativity

Galilean transformations are associative under composition. The Galilei group is not abelian.

Student work – proof

Prove associativity and exhibit one pair of Galilean transformations that does not commute.

3.3 Why this is a Lie group**Definition 3.6: Lie group**

A *Lie group* is a group that is also a smooth manifold, such that multiplication $(g_2, g_1) \mapsto g_2 \circ g_1$ and inversion $g \mapsto g^{-1}$ are smooth maps.

A smooth manifold is a space that can locally be described by real coordinates with smooth changes between overlapping coordinate systems. For a Lie group, both composition and inversion must depend smoothly on these local coordinates. Merely having continuously varying parameters is not sufficient.

A Galilean transformation is parametrized by one time translation, three spatial translations, three boost parameters, and three rotation parameters. Its dimension is therefore

$$\dim \text{Gal}(3) = 1 + 3 + 3 + 3 = 10.$$

More precisely, the parameter space has the manifold structure $\mathbb{R} \times \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3)$, and the composition and inverse formulas are smooth. The Galilei group is consequently a ten-dimensional Lie group. In particular, $SO(3)$ has three local parameters, although it is not globally the same manifold as $\mathbb{R}^3$. The displayed product describes the parameter *manifold*, not a direct-product group law.

Quantum-mechanical remark

In quantum mechanics, state vectors differing only by an overall phase represent the same physical state. Consequently, the operators representing Galilean transformations may compose only up to such a phase,

$$U(g_2)U(g_1) = e^{i\theta(g_2, g_1)}U(g_2 \circ g_1), \quad \theta(g_2, g_1) \in \mathbb{R}.$$

This is called a *projective representation*. It concerns the action on quantum states, not the coordinate transformations studied here, and is not needed in the rest of the chapter.

4 Why Galilean invariance fails for light

SOURCE SLIDE 4

The Galilei group preserves Newton's law of inertial motion, but a successful kinematics must also be compatible with electromagnetism. In vacuum, Maxwell's equations imply electromagnetic waves with speed

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}.$$

The constants ε_0 and μ_0 describe the electromagnetic vacuum rather than the motion of a particular source. The empirical statement that inertial observers obtain the same value of c is therefore in direct tension with Galilean velocity addition.

To see the conflict, it is convenient to use the conventional physical-velocity form of a Galilean boost along the x -axis. It is

$$t' = t, \quad x' = x - Vt.$$

For a light ray satisfying $x = ct$, this transformation gives $x' = (c - V)t$ and hence $c' = c - V$. The measured speed would depend on the observer.

Proposition 4.1: The vacuum wave equation is not Galilean invariant

For a scalar wave amplitude ϕ , the one-dimensional vacuum wave equation

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \frac{\partial^2 \phi}{\partial x^2} = 0$$

does not retain the same form under $t' = t$, $x' = x - Vt$ when $V \neq 0$, with $\phi'(t', x') = \phi(t, x)$.

Student work – derivation

Prove Proposition 4.1.

Key idea. The conflict cannot be repaired while retaining Galilean time. The relation between inertial coordinates must transform time together with space. The resulting coordinate changes are Lorentz transformations.

5 The postulates of special relativity

SOURCE SLIDE 5

Postulate 5.1: Relativity principle

The laws of physics have the same mathematical form in every inertial frame. No experiment performed entirely inside an inertial laboratory can identify a preferred inertial state of uniform motion.

Postulate 5.2: Invariant speed of light

Light in vacuum propagates with the same speed c in every inertial frame, independently of the motion of the source.

The standard derivation also assumes homogeneity of space and time, isotropy of space, and smooth dependence of frame transformations on relative velocity. Inertial frames use equally calibrated rods and clocks, with Einstein synchronization by light signals. In this convention, a reflected light signal is assigned a reflection time halfway between its emission and reception times on the sending clock. Homogeneity and the preservation of inertial motion lead to affine coordinate changes. Combined with the postulates and these conventions, their linear parts are Lorentz transformations rather than Galilean transformations. Nonzero boosts therefore mix spatial and temporal coordinates.

What follows, and what does not

The resulting transformations preserve the light cone and distinguish timelike, lightlike, and spacelike separations. Causal influences are assumed to propagate within or on the future light cone. This excludes signals faster than c . The geometric transformation laws alone do not specify which signals or particles exist.

A *field* assigns a physical quantity to each event. A scalar field assigns a number, while a vector field assigns a vector. Spinor fields use a different transformation rule and occur, for example, in the description of electrons. These are possible types of fields, not predictions of the two postulates. Spinor theory is not required here. Scalar, vector, and tensor fields are introduced near the end of the chapter.

Student work – conceptual argument

Explain why Galilean absolute time is incompatible with the same finite, nonzero light speed in every inertial frame.

6 Points, vectors, and coordinates

SOURCE SLIDE 6

6.1 The affine plane

The spacetime discussion has already required a distinction between an event and its coordinates. The same distinction appears in ordinary Euclidean geometry, where a point is not itself a vector. Affine spaces provide the natural language for making this separation precise.

Definition 6.1: Affine space

An *affine space* modeled on a vector space V is a nonempty set A equipped with a map

$$A \times V \longrightarrow A, \quad (P, \mathbf{v}) \longmapsto P + \mathbf{v},$$

satisfying $P + \mathbf{0} = P$ and $(P + \mathbf{v}) + \mathbf{w} = P + (\mathbf{v} + \mathbf{w})$. For every $P, Q \in A$, there is a unique $\mathbf{v} \in V$ such that $Q = P + \mathbf{v}$. This vector is denoted $Q - P$. No point of A is preferred as an origin.

The Euclidean plane is a two-dimensional affine space modeled on $V = \mathbb{R}^2$, together with a positive-definite inner product on its displacement vectors. Once an origin O and a basis $(\mathbf{e}_1, \mathbf{e}_2)$ have been chosen, a point P is represented by the position vector

$$\overrightarrow{OP} = x^1 \mathbf{e}_1 + x^2 \mathbf{e}_2.$$

Thus, for any fixed point P_0 , the whole affine plane is

$$A = \{P_0 + a\mathbf{e}_1 + b\mathbf{e}_2 \mid a, b \in \mathbb{R}\}.$$

Remark

A point is not intrinsically a vector. Only the difference of two points is a vector without further choices. A position vector appears after selecting an origin, while Euclidean lengths and angles appear only after selecting an inner product.

6.2 Bases and coordinate systems

Definition 6.2: Basis and span

The *span* of vectors $\mathbf{e}_1, \mathbf{e}_2$ is the set of their linear combinations. They form a *basis* of a two-dimensional vector space V when they are linearly independent and span V . Every $\mathbf{v} \in V$ then has a unique expansion

$$\mathbf{v} = v^1 \mathbf{e}_1 + v^2 \mathbf{e}_2.$$

The coefficients (v^1, v^2) are the components of $\mathbf{v}$ in this basis.

Affine coordinates use an origin and a fixed basis. When that basis is orthonormal, they are Cartesian coordinates. Polar coordinates (r, θ) instead label points through the relations

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Because this relation is nonlinear, changing from Cartesian to polar coordinates is not a change of a fixed vector-space basis. Polar coordinates also require a choice of angular range and are singular at $r = 0$.

Proposition 6.3: Existence and uniqueness of components

An ordered pair $(\mathbf{e}_1, \mathbf{e}_2)$ gives every vector in V a unique pair of components if and only if it is a basis of V .

Student work – proof

Prove Proposition 6.3.

7 Products of vectors

SOURCE SLIDE 7

Scalar multiplication has an unambiguous meaning because the vector-space axioms already define $\lambda \mathbf{a}$ for $\lambda \in \mathbb{R}$. By contrast, there is no universal product of two vectors. Different products encode different geometric information and may even take values in different spaces.

7.1 Inner and tensor products

In an orthonormal Cartesian basis of the Euclidean plane,

$$\mathbf{a} \cdot \mathbf{b} = a^1 b^1 + a^2 b^2.$$

The inner product returns a scalar and depends on the Euclidean metric. The tensor product instead produces an element of $V \otimes V$, written

$$\mathbf{a} \otimes \mathbf{b}, \quad (\mathbf{a} \otimes \mathbf{b})^{ij} = a^i b^j.$$

Unlike the inner product, this construction does not require a metric. The space $V \otimes V$ is generated by symbols $\mathbf{a} \otimes \mathbf{b}$ subject to bilinearity in $\mathbf{a}$ and $\mathbf{b}$. Its general elements are sums of such products, not necessarily single products. In a basis $(\mathbf{e}_i)$, the products $\mathbf{e}_i \otimes \mathbf{e}_j$ form a basis of $V \otimes V$.

Student work – calculation

For $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (-1, 3)$, calculate $\mathbf{a} \cdot \mathbf{b}$ and the component array of $\mathbf{a} \otimes \mathbf{b}$.

7.2 The geometric product

This optional construction combines two kinds of geometric information in one multiplication. An *algebra* here is a real vector space equipped with a bilinear product. Scalars and vectors can belong to the same algebra without being the same kind of element.

Definition 7.1: Euclidean geometric algebra

Let V be a finite-dimensional Euclidean vector space with inner product g . Its *geometric algebra*, denoted by $\text{Cl}(V, g)$, is the unital associative real algebra generated by V , subject only to bilinearity and the relations

$$\mathbf{a}^2 = \|\mathbf{a}\|^2 1$$

for every $\mathbf{a} \in V$, where 1 is the multiplicative identity. This multiplication is called the *geometric product*. Scalar multiples of 1 are identified with real numbers, so the identity is usually omitted when writing a scalar. The product is generally not commutative.

For two vectors, the geometric product separates naturally into a symmetric part and an antisymmetric part,

$$\mathbf{a}\mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}.$$

The two parts are recovered from

$$\mathbf{a} \cdot \mathbf{b} = \frac{1}{2}(\mathbf{a}\mathbf{b} + \mathbf{b}\mathbf{a}), \quad \mathbf{a} \wedge \mathbf{b} = \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a}).$$

For nonzero vectors, the scalar part is $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$. Dividing it by $\|\mathbf{b}\|$ gives the signed scalar projection of $\mathbf{a}$ along $\mathbf{b}$. By contrast, $\mathbf{a} \wedge \mathbf{b}$ represents the oriented parallelogram spanned by the two vectors. Its magnitude is

$$\|\mathbf{a} \wedge \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta,$$

where $0 \leq \theta \leq \pi$ is the angle between the nonzero vectors. The wedge product is bilinear and alternating,

$$\mathbf{b} \wedge \mathbf{a} = -\mathbf{a} \wedge \mathbf{b}, \quad \mathbf{a} \wedge \mathbf{a} = 0.$$

In particular, $\mathbf{a} \wedge \mathbf{b} = 0$ whenever the vectors are linearly dependent, including when either vector is zero.

Definition 7.2: Bivectors and multivectors

A *bivector* is a linear combination of exterior products $\mathbf{a} \wedge \mathbf{b}$. A single such product is called *simple* and, when nonzero, represents an oriented area element. In dimensions four and higher, a general bivector need not represent just one plane.

A *multivector* is a sum of objects of different grades. In an n -dimensional geometric algebra it has the form

$$A = \langle A \rangle_0 + \langle A \rangle_1 + \cdots + \langle A \rangle_n,$$

where $\langle A \rangle_0$ is the scalar part, $\langle A \rangle_1$ is the vector part, and $\langle A \rangle_2$ is the bivector part. More generally, the grade- k part is a linear combination of exterior products of k vectors. Grade zero consists of scalars.

For example, if $\mathbf{e}_1$ and $\mathbf{e}_2$ are orthonormal, then

$$\mathbf{e}_1 \cdot \mathbf{e}_2 = 0, \quad \mathbf{e}_1 \mathbf{e}_2 = \mathbf{e}_1 \wedge \mathbf{e}_2, \quad \mathbf{e}_2 \mathbf{e}_1 = -\mathbf{e}_1 \mathbf{e}_2.$$

Thus the geometric product retains both the metric information contained in the inner product and the oriented-plane information contained in the exterior product.

Geometric algebra and tensors

The notation $\bigwedge^k V$, read as the k -th exterior power of V , denotes the space of linear combinations of k -fold wedge products of vectors. It is naturally identified with the alternating part of $V^{\otimes k}$. Its elements are called k -vectors and may therefore be represented as antisymmetric *contravariant* tensors of order k .

The dual space V^* consists of linear maps $\alpha : V \rightarrow \mathbb{R}$, called covectors. An element of $\bigwedge^k V^*$ is an alternating k -linear map

$$\omega : V \times \cdots \times V \longrightarrow \mathbb{R}$$

and is called a k -form. Alternating means that interchanging two arguments reverses the sign. A k -form acts on a k -vector and returns a number. For example, a two-form can measure an oriented area element.

A Euclidean metric g allows a vector $\mathbf{v}$ to be associated with the covector

$$v^b(\mathbf{w}) = g(\mathbf{v}, \mathbf{w}).$$

This induces an identification of $\bigwedge^k V$ with $\bigwedge^k V^*$, but it depends on the chosen metric. The two spaces are therefore related, but their elements are not intrinsically the same objects.

A k -form is therefore an alternating *covariant* tensor. General tensors need not be alternating, and a multivector may combine several tensor orders. The essential distinction is between the geometric product and the tensor product, not a blanket assertion that multivectors are unrelated to tensors. Geometric algebra will not be needed later in the chapter.

Student work – derivation

Derive the symmetric and antisymmetric parts of the geometric product.

8 What is a tensor?

SOURCE SLIDE 8

Vectors belong to a larger family of multilinear objects called tensors. The dual space supplies the second ingredient needed to define this family, independently of the optional geometric-algebra discussion.

Definition 8.1: Dual space and covectors

For a real vector space V , the *dual space* V^* is the vector space of linear maps $\omega : V \rightarrow \mathbb{R}$, called *covectors*. If $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is a basis of V , its dual basis $(\varepsilon^1, \dots, \varepsilon^n)$ is characterized by

$$\varepsilon^i(\mathbf{e}_j) = \delta^i_j,$$

where δ^i_j equals one when $i = j$ and zero otherwise. Thus ε^i extracts the i -th component of a vector.

A covector is a linear measuring rule, not a vector written sideways. A metric can associate a vector with a covector, but the definition of V^* does not require a metric.

Definition 8.2: Tensor of type (r, s)

Let V be a finite-dimensional vector space and V^* its dual. A tensor of type (r, s) is an element of

$$V^{\otimes r} \otimes (V^*)^{\otimes s}.$$

Equivalently, it is a multilinear map

$$T : \underbrace{V^* \times \dots \times V^*}_{r \text{ factors}} \times \underbrace{V \times \dots \times V}_{s \text{ factors}} \longrightarrow \mathbb{R}.$$

Its tensor order is $r + s$, where r, s are nonnegative integers. A tensor of type $(0, 0)$ is a scalar.

In a chosen basis, a tensor is represented by components

$$T^{i_1 \dots i_r}_{j_1 \dots j_s}.$$

Changing the basis changes this array according to a definite transformation law. The tensor is the geometric object represented by all such arrays, not a particular array and not merely an expression carrying indices.

For example, a vector is of type $(1, 0)$, a covector is of type $(0, 1)$, and a bilinear form $g(v, w)$ is of type $(0, 2)$. The tensor product $\mathbf{a} \otimes \mathbf{b}$ is of type $(2, 0)$. It acts on two covectors by

$$(\mathbf{a} \otimes \mathbf{b})(\alpha, \beta) = \alpha(\mathbf{a}) \beta(\mathbf{b}).$$

Multilinearity means linearity in each argument separately while the other arguments are held fixed.

Two meanings of rank

In much of the relativity literature, “rank-two tensor” means a tensor with two indices, more precisely a tensor of order two. Matrix rank is a different concept. Moreover, an array of numbers becomes the component array of a tensor only after its geometric role and basis-change law have been specified.

For fixed (r, s) , tensors form a vector space. Addition is represented by

$$(S + T)^{i_1 \dots i_r}_{j_1 \dots j_s} = S^{i_1 \dots i_r}_{j_1 \dots j_s} + T^{i_1 \dots i_r}_{j_1 \dots j_s}.$$

Scalar multiplication is represented componentwise in the same way.

Student work – proof

Prove that tensors of a fixed type (r, s) form a vector space.

9 Contraction and Einstein notation

SOURCE SLIDE 9

9.1 Composition as contraction

The tensor product can be followed by a contraction, in which one contravariant and one covariant slot are paired. Matrix multiplication is the familiar example. If $A : V \rightarrow V$ and $B : V \rightarrow V$ have components A^i_j and B^j_k , then $A \circ B$ has components

$$(A \circ B)^i_k = \sum_{j=1}^n A^i_j B^j_k.$$

The repeated index j is *contracted*. The free indices i, k identify the type of the result.

Definition 9.1: Einstein summation convention

Within the Einstein summation convention, an index repeated exactly twice in one term, once upstairs and once downstairs, is summed over its full range. For example,

$$A^i{}_j B^j{}_k := \sum_{j=1}^n A^i{}_j B^j{}_k.$$

Every term in a valid tensor equation must have the same free indices in the same positions.

Student work – index calculation

Write $C^i{}_k = A^i{}_j B^j{}_k$ explicitly for $n = 2$, component by component.

9.2 Vectors and covectors

A vector $v \in V$ has contravariant components v^i . A covector $\omega \in V^*$ has covariant components ω_i and acts on v by

$$\omega(v) = \omega_i v^i.$$

An inner product defines an isomorphism between vectors and covectors. In an orthonormal Euclidean basis this identification makes a covector's row of components look like the transpose of a vector's column. The resemblance is a consequence of the metric, not a definition of a covector.

Remark

The position of an index does not specify a row or column on the page. Both vectors and covectors can be stored as columns, but their transformation matrices then differ. The formula for lowering an index is $v_i = g_{ij} v^j$, not transposition of an individual component. In an orthonormal Euclidean basis, $g_{ij} = \delta_{ij}$, so the upper and lower components happen to have the same numerical values.

10 Four-vector notation and units

SOURCE SLIDE 10

Spatial indices i, j, k will range over $1, 2, 3$, while Greek spacetime indices μ, ν, ρ will range over $0, 1, 2, 3$. In an inertial frame, an event X receives coordinates

$$x^\mu(X) = (ct, x, y, z).$$

This coordinate tuple is often called the four-position, although it depends on the chosen origin. Its difference Δx^μ for two events is the component array of a genuine displacement vector. A momentum four-vector is written without a spatial arrow as

$$p^\mu = \left(\frac{E}{c}, p^1, p^2, p^3 \right) = \left(\frac{E}{c}, \mathbf{p} \right).$$

Every entry in the event-coordinate tuple has units of length, and every component of p^μ has units of momentum.

From this point onward, natural units are used unless c is displayed explicitly. In these units

$$c = 1$$

and the same coordinates and momentum components are written

$$x^\mu = (t, \mathbf{x}), \quad p^\mu = (E, \mathbf{p}).$$

What $c = 1$ means

Setting $c = 1$ chooses a common unit for time and length. It does not assert that a dimensionful SI quantity is literally the pure number one. The useful conversions are

$$1 \text{ s} \longleftrightarrow 2.99792458 \times 10^8 \text{ m}, \quad 1 \text{ m} \longleftrightarrow 3.33564 \text{ ns}.$$

Student work – unit conversion

Express $2.0 \mu\text{s}$ as a distance and 15 m as a time when $c = 1$.

11 Invariants and Minkowski geometry

SOURCE SLIDE 11

The preceding sections distinguished geometric objects from their coordinate representations. The next question is which quantities constructed from these objects remain unchanged under a change of inertial frame.

11.1 Euclidean invariants

Definition 11.1: Invariant

Relative to a specified family of transformations, an *invariant* is a quantity whose value is unchanged by every transformation in that family. Coordinate formulas may change while the invariant geometric value remains the same.

In Euclidean space, rigid translations and rotations preserve the squared separation

$$\ell^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2.$$

They also preserve the angle between nonzero vectors determined by

$$\cos \alpha = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

These are invariants of Euclidean geometry.

11.2 The Minkowski metric

Special relativity retains affine spacetime but replaces absolute Galilean time and the separate Euclidean spatial metric with a single spacetime metric.

Definition 11.2: Minkowski metric and scalar product

The *Minkowski metric* is a symmetric, nondegenerate bilinear form on the spacetime translation space $\mathcal{V}$ with signature $(+, -, -, -)$. In an inertial basis its component matrix is

$$(g_{\mu\nu}) = \text{diag}(1, -1, -1, -1).$$

For four-vectors p and q , their Minkowski scalar product is

$$p \cdot q := g_{\mu\nu} p^\mu q^\nu = p^0 q^0 - p^1 q^1 - p^2 q^2 - p^3 q^3.$$

Nondegeneracy means that if $p \cdot q = 0$ for *every* q , then $p = 0$. It does not mean that $p \cdot p = 0$ forces $p = 0$. This distinction is essential for a metric with both positive and negative signs.

For events A and B , the coordinate components of their displacement are $\Delta x^\mu = x^\mu(B) - x^\mu(A)$. In units $c = 1$, its squared interval is

$$\Delta s^2 = g_{\mu\nu} \Delta x^\mu \Delta x^\nu = (\Delta t)^2 - \|\Delta \mathbf{x}\|^2.$$

With c restored, the first term is $c^2(\Delta t)^2$.

Remark

The notation Δs^2 does not imply non-negativity. A nonzero vector can have a positive, negative, or even zero Minkowski square because the metric is indefinite. Thus $\sqrt{\Delta s^2}$ is not a real-valued norm on all spacetime displacements.

Student work – calculation

For $\Delta x^\mu = (5, 3, 0, 0)$ and $\Delta y^\mu = (2, -1, 1, 0)$, calculate $\Delta x \cdot \Delta x$, $\Delta y \cdot \Delta y$, and $\Delta x \cdot \Delta y$.

12 Events, intervals, and causal type

SOURCE SLIDE 12–13

12.1 Coordinate dependence and invariant separation

Let $A, B \in \mathcal{M}$ be events. In one inertial frame, write $\chi_O(A) = a = (a^0, a^1, a^2, a^3)$ and $\chi_O(B) = b = (b^0, b^1, b^2, b^3)$. The geometric displacement $d = B - A$ is represented in that frame by the components

$$d^\mu = b^\mu - a^\mu.$$

Another inertial frame assigns different components to the same displacement. Nevertheless, its Minkowski square

$$d^2 := d \cdot d = g_{\mu\nu} d^\mu d^\nu$$

has the same value. Translations cancel in $b^\mu - a^\mu$, while Lorentz transformations preserve the metric. The scalar d^2 therefore depends on the two events, not on the inertial coordinates used to describe them.

Key idea. An invariant can be evaluated in any inertial frame. One should therefore choose a frame in which the relevant components are simplest.

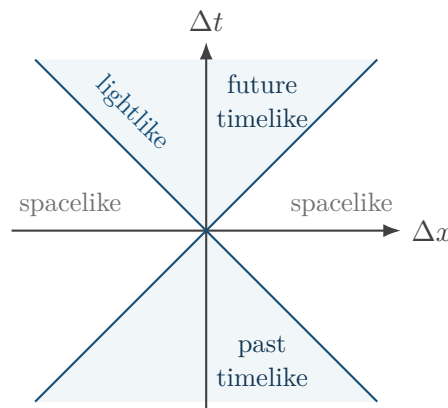
12.2 Classification

Definition 12.1: Causal type

A nonzero displacement $d \in \mathcal{V}$, with components d^μ , is

timelike	if $d^2 > 0$,
null or lightlike	if $d^2 = 0$,
spacelike	if $d^2 < 0$.

After a future time direction has been chosen, a timelike or nonzero null displacement is *future-directed* when $d^0 > 0$ and *past-directed* when $d^0 < 0$. An *orthochronous* Lorentz transformation preserves this distinction. A *proper* Lorentz transformation has determinant +1. The group terminology is developed further in Section 14.



A 1 + 1-dimensional section with $\Delta y = \Delta z = 0$ and $c = 1$.

Theorem 12.2: Canonical forms of a displacement

After a spatial rotation and a proper orthochronous Lorentz boost, a nonzero displacement has one of the following canonical forms.

1. If $d^2 > 0$, there is a frame in which

$$d^\mu = (\pm\sqrt{d^2}, 0, 0, 0),$$

the sign is positive for a future-directed displacement and negative for a past-directed one.

2. If $d^2 = 0$ and $d^\mu \neq 0$, there is no frame in which the events are simultaneous or colocated, and a suitable orientation gives

$$d^\mu = (\lambda, \lambda, 0, 0), \quad \lambda \neq 0,$$

with the time orientation fixing the sign of λ . Its magnitude can be changed by a boost and is not an invariant.

3. If $d^2 < 0$, there is a frame in which

$$d^\mu = (0, \sqrt{-d^2}, 0, 0).$$

The three signs correspond to three distinct causal possibilities. Timelike separated events can occur at the same place in some inertial frame. Spacelike separated events can occur at the same time in some inertial frame, and their temporal order can be reversed by changing frames. A nonzero null displacement admits neither a common-place frame nor a simultaneous frame.

Time ordering

For timelike or distinct lightlike separated events, their temporal order is the same in every orthochronous inertial frame. Spatial reflection does not alter this conclusion, whereas time reversal does. For spacelike separated events, neither temporal ordering is shared by all such frames.

Student work – proof

Prove Theorem 12.2.

12.3 Intervals and clock readings

For a timelike separation, the canonical frame contains an inertial clock that passes through both events. The elapsed time on that clock is $\sqrt{\Delta s^2}/c$. The qualification *inertial* matters because the reading of a clock on an accelerated path depends on the path, not just its endpoints.

Definition 12.3: Proper time

Along a future-directed timelike worldline, the *proper time* τ is the time measured by an ideal clock following that worldline. Its infinitesimal increment is

$$d\tau = \frac{1}{c} \sqrt{g_{\mu\nu} dx^\mu dx^\nu} = \sqrt{1 - \frac{\|\mathbf{u}\|^2}{c^2}} dt, \quad \mathbf{u} = \frac{d\mathbf{x}}{dt},$$

where $x^0 = ct$, $dt > 0$, and $\|\mathbf{u}\| < c$. The elapsed proper time is the integral of this expression along the path.

For spacelike separated events, $\sqrt{-\Delta s^2}$ is their spatial separation in a frame in which they are simultaneous. A nonzero null worldline has no inertial rest frame. Thus a null interval must not be interpreted as the reading of a clock in a photon's rest frame.

13 Four-momentum and invariant mass

SOURCE SLIDE 14

Momentum and energy combine into one Lorentz-covariant object. In natural units, the energy-momentum four-vector is

$$p^\mu = (E, \mathbf{p}).$$

For a massive particle, a rest frame exists in which $\mathbf{p} = \mathbf{0}$. The rest mass m is defined by the rest-frame energy $E_{\text{rest}} = m$ in natural units. Since $p^2 = p \cdot p$ is invariant,

$$p^2 = E^2 - \|\mathbf{p}\|^2 = m^2.$$

Restoring c gives

$$E^2 = \|\mathbf{p}\|^2 c^2 + m^2 c^4.$$

Massless particles

A massless particle has no rest frame. Its four-momentum is null, so

$$p^2 = 0, \quad E = c\|\mathbf{p}\|$$

for a positive-energy particle. The rest-frame argument applies only when $m > 0$. With c explicit, the invariant relation is $p^2 = m^2 c^2$. In units $c = 1$, it is $p^2 = m^2$, including the massless case.

Student work – derivation

Derive the energy-momentum relation and restore the factors of c .

For an isolated collision, four-momentum conservation states

$$P_{\text{in}}^\mu = \sum_{i \in \text{in}} p_i^\mu = \sum_{j \in \text{out}} p_j^\mu = P_{\text{out}}^\mu.$$

Call this common total four-momentum P . Conservation compares its value before and after the collision, whereas Lorentz invariance compares P^2 between frames. These are different statements. In units $c = 1$, the Mandelstam variable s for two incoming particles is

$$s = (p_1 + p_2)^2.$$

If P is timelike and future-directed, a center-of-momentum frame exists. There $\mathbf{P} = \mathbf{0}$, so $\sqrt{s}$ is the total center-of-momentum energy. The symbol s here is a momentum invariant, not the spacetime interval Δs^2 .

Student work – calculation

Expand $s = (p_1 + p_2)^2$ in terms of m_1, m_2 , and $p_1 \cdot p_2$.

14 Lorentz and Poincaré transformations

SOURCE SLIDE 15

The invariant interval now determines the allowed transformations between inertial coordinate systems. Translations alter the chosen spacetime origin, while the linear part must preserve the Minkowski metric.

Definition 14.1: Lorentz transformation

A linear map $\Lambda : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ is a *Lorentz transformation* if it preserves the Minkowski scalar product. Thus

$$(\Lambda p) \cdot (\Lambda q) = p \cdot q \quad \text{for all } p, q.$$

In matrix notation this is equivalent to

$$\Lambda^\top g \Lambda = g.$$

All Lorentz transformations form the group $O(1, 3)$, where 1, 3 count the positive and negative signs of the metric. The Lorentz condition implies $\det \Lambda = \pm 1$ and $|\Lambda^0_0| \geq 1$. These two sign choices distinguish four connected components. Here a connected component is a maximal family within which matrices can be joined by continuous paths that remain Lorentz transformations.

The component containing the identity is the *proper orthochronous* Lorentz group $SO^+(1, 3)$, characterized by

$$\det \Lambda = +1, \quad \Lambda^0_0 \geq 1.$$

It contains ordinary spatial rotations and boosts. Time reversal and spatial reflection are also in $O(1, 3)$, but not in this component. The adjective *proper* by itself does not exclude time reversal combined with spatial reflection.

Definition 14.2: Poincaré transformation

An affine spacetime transformation

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu$$

with Lorentz matrix Λ is a *Poincaré transformation*. The special case $a^\mu = 0$ is a homogeneous Lorentz transformation.

Remark

The expression *inhomogeneous Lorentz transformation* is sometimes used for (Λ, a) . The standard name for the complete symmetry group is the Poincaré group. Translations do not change event displacements, so preservation of intervals reduces to the Lorentz condition on Λ .

The Lorentz group has six continuous parameters near the identity, consisting of three rotation parameters and three boost parameters. Adding four translations makes the Poincaré group ten-dimensional.

Student work – derivation

Derive the matrix condition for a Lorentz transformation and determine the number of independent infinitesimal parameters.

15 The standard boost

SOURCE SLIDE 16

A proper orthochronous Lorentz transformation can be built from rotations and boosts. The other components also require a discrete reflection or time reversal. To isolate the simplest boost, let O' move with signed physical velocity V along the x -axis relative to O . Choose coincident origins at $t = t' = 0$ and parallel spatial axes. Symmetry then gives $y' = y$ and $z' = z$, so only $x^0 = t$ and $x^1 = x$ mix in units $c = 1$.

The full four-dimensional matrix, denoted by $\Lambda(V)$, has block form

$$\Lambda(V) = \begin{pmatrix} \Lambda^0_0 & \Lambda^0_1 & 0 & 0 \\ \Lambda^1_0 & \Lambda^1_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The unchanged transverse coordinates already preserve their contribution $-(\Delta y)^2 - (\Delta z)^2$ to the interval. The metric restricted to the (t, x) -plane is represented by

$$g_{(1+1)} := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The notation $1+1$ indicates one time and one spatial dimension. The remaining problem is therefore to find its 2×2 boost block L , which must preserve this restricted metric. Thus $\Lambda(V) = \text{diag}(L, \mathbf{I}_2)$. Here L and Λ denote the block and the full matrix, respectively.

Student work – derivation

Obtain the independent conditions on the four entries of the boost block.

16 Determining the boost matrix

SOURCE SLIDE 17

Write the unknown $1+1$ -dimensional block as

$$L = \begin{pmatrix} \ell_{00} & \ell_{01} \\ \ell_{10} & \ell_{11} \end{pmatrix}.$$

The subscripts on ℓ_{ab} label matrix positions only. They are not lowered Lorentz indices. The Lorentz condition is

$$L^\top g_{(1+1)} L = g_{(1+1)}.$$

Theorem 16.1: Boost connected to the identity

Every real 2×2 matrix satisfying $L^\top g_{(1+1)} L = g_{(1+1)}$ in the component containing the identity can be written uniquely as

$$L(\eta) = \begin{pmatrix} \cosh \eta & -\sinh \eta \\ -\sinh \eta & \cosh \eta \end{pmatrix},$$

where $\eta \in \mathbb{R}$ is the rapidity.

Remark

The group $O(1, 1)$ has four disconnected components. The standard boosts $L(\eta)$ form the component containing the identity because $L(0) = \mathbf{I}_2$ and varying η continuously connects each boost to $\mathbf{I}_2$.

The other components are obtained by combining $L(\eta)$ with spatial reflection $P = \text{diag}(1, -1)$, time reversal $T = \text{diag}(-1, 1)$, or both. They therefore have the forms

$$PL(\eta), \quad TL(\eta), \quad PTL(\eta).$$

In these products the boost acts first. The matrices P and T then reverse the resulting spatial and temporal coordinates, respectively. The signs of the matrix entries are constrained by the Lorentz condition and cannot be chosen independently.

Student work – proof

Prove Theorem 16.1.

17 Rapidity and relative velocity

SOURCE SLIDE 18

The rapidity parameter acquires physical meaning by following the primed origin. Let A and B be two events on its worldline, with A chosen as the common origin. The displacement from A to B has components

$$\Delta x^\mu = (\Delta t, V\Delta t, 0, 0) \quad \text{in } O, \quad \Delta x'^\mu = (\Delta t', 0, 0, 0) \quad \text{in } O'.$$

Applying the boost of Theorem 16.1 gives

$$\begin{pmatrix} \Delta t' \\ 0 \end{pmatrix} = \begin{pmatrix} \cosh \eta & -\sinh \eta \\ -\sinh \eta & \cosh \eta \end{pmatrix} \begin{pmatrix} \Delta t \\ V\Delta t \end{pmatrix}.$$

Proposition 17.1: Velocity, rapidity, and the Lorentz factor

For a standard boost with $|V| < c$, define $\beta = V/c$. Then

$$\beta = \tanh \eta, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}} = \cosh \eta, \quad \gamma\beta = \sinh \eta.$$

In units $c = 1$, V and β have the same numerical value. With c displayed explicitly, the standard boost is

$$\begin{aligned} ct' &= \gamma(ct - \beta x), \\ x' &= \gamma(x - \beta ct), \\ y' &= y, \quad z' = z. \end{aligned}$$

Student work – derivation

Derive Proposition 17.1.

The nonrelativistic limit is controlled by the dimensionless velocity $\beta = V/c$. When $|\beta| \ll 1$,

$$\eta = \operatorname{artanh} \beta = \beta + \mathcal{O}(\beta^3), \quad \gamma = 1 + \frac{1}{2}\beta^2 + \mathcal{O}(\beta^4).$$

Equivalently, the boost block has the expansion

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \left[\mathbf{I}_2 - \beta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \frac{\beta^2}{2} \mathbf{I}_2 + \mathcal{O}(\beta^3) \right] \begin{pmatrix} ct \\ x \end{pmatrix}.$$

Keeping the terms linear in β gives $x' \simeq x - Vt$ and $t' \simeq t - Vx/c^2$. The Galilean limit is more precisely obtained by letting $c \rightarrow \infty$ at fixed V, x, t . Then $\gamma \rightarrow 1$, $t' \rightarrow t$, and $x' \rightarrow x - Vt$.

Proposition 17.2: Composition of collinear boosts

For collinear boosts,

$$L(\eta_2)L(\eta_1) = L(\eta_1 + \eta_2).$$

Velocities do not add linearly, but rapidities do. If V_1 is the velocity of O' in O and V_2 is the velocity of O'' in O' , with parallel axes, the velocity of O'' in O is

$$V_{\text{tot}} = \frac{V_1 + V_2}{1 + V_1 V_2 / c^2}.$$

Student work – proof

Prove both the rapidity and velocity composition laws in Proposition 17.2.

18 Four-vectors and covectors

SOURCE SLIDE 19

A spacetime displacement is the difference of two events in the affine spacetime $\mathcal{M}$. Because translations cancel, its components transform homogeneously according to

$$\Delta x'^\mu = \Lambda^\mu{}_\nu \Delta x^\nu.$$

A *four-vector* V has components obeying the same linear law, $V'^\mu = \Lambda^\mu{}_\nu V^\nu$, although it need not physically be a displacement. Four-momentum, for example, has units of momentum. A *covector* is a linear functional on four-vectors. If its components are W_μ , the pairing $W_\mu V^\mu$ is a scalar and must be independent of the inertial basis.

Proposition 18.1: Covector transformation law

If $V'^\mu = \Lambda^\mu{}_\nu V^\nu$, scalar invariance requires

$$W'_\mu = (\Lambda^{-1})^\nu{}_\mu W_\nu.$$

Student work – proof

Prove Proposition 18.1.

For the standard boost,

$$L(\eta)^{-1} = L(-\eta), \quad \Lambda(V)^{-1} = \Lambda(-V).$$

This has a direct physical meaning. The velocity of O as measured in O' is the negative of the velocity of O' as measured in O .

Student work – verification

Verify $L(\eta)^{-1} = L(-\eta)$.

19 The metric and index operations

SOURCE SLIDE 20

The metric not only produces invariant scalars. It also identifies vectors with covectors. In the standard inertial basis, the metric and inverse metric have components

$$(g_{\mu\nu}) = \text{diag}(1, -1, -1, -1), \quad (g^{\mu\nu}) = \text{diag}(1, -1, -1, -1),$$

and satisfy

$$g^{\mu\nu} g_{\nu\rho} = \delta^\mu{}_\rho.$$

They lower and raise indices according to

$$V_\mu = g_{\mu\nu} V^\nu, \quad V^\mu = g^{\mu\nu} V_\nu.$$

Thus, if $V^\mu = (V^0, V^1, V^2, V^3)$, then

$$V_\mu = (V^0, -V^1, -V^2, -V^3).$$

Proposition 19.1: Inverse of a Lorentz matrix

The Lorentz condition implies

$$\Lambda^{-1} = g^{-1} \Lambda^\top g.$$

In the standard inertial basis, $g^{-1} = g$, so this is also $\Lambda^{-1} = g \Lambda^\top g$. In components, define

$$\Lambda_\mu{}^\nu := g_{\mu\alpha} \Lambda^\alpha{}_\beta g^{\beta\nu}.$$

Then

$$(\Lambda^{-1})^\mu{}_\nu = \Lambda_\nu{}^\mu.$$

Student work – proof

Prove Proposition 19.1.

Combining the metric with the vector transformation gives

$$V'_\mu = (\Lambda^{-1})^\nu{}_\mu V_\nu = \Lambda_\mu{}^\nu V_\nu.$$

The upper-index and lower-index formulas therefore encode the vector and dual representations of the Lorentz group. In matrix notation, if the components of both objects are stored as columns, the vector column transforms with Λ , whereas the covector column transforms with $(\Lambda^{-1})^\top$. Writing a covector as a row instead gives right multiplication by Λ^{-1} .

Student work – derivation

Derive the lower-index transformation law.

20 Lorentz transformations of fields

SOURCE SLIDE 21

A field assigns a quantity to every event in a region of spacetime. The type of that quantity determines the field's transformation law. The fields below are taken to be smooth, so that derivatives can be formed when needed. No field equation is assumed.

20.1 One event, two coordinate descriptions

Let $X \in \mathcal{M}$ be one event. Inertial frames O and O' assign it coordinates $x = \chi_O(X)$ and $x' = \chi_{O'}(X)$, related by

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu.$$

The argument displayed in a component function is therefore a coordinate label for X , not a second event.

Definition 20.1: Scalar field transformation

A geometric scalar field is a map $\Phi : \mathcal{M} \rightarrow \mathbb{R}$. Its coordinate representatives $\phi = \Phi \circ \chi_O^{-1}$ and $\phi' = \Phi \circ \chi_{O'}^{-1}$ satisfy

$$\boxed{\phi'(x') = \phi(x)}.$$

Definition 20.2: Vector and covector field transformations

A vector field assigns a spacetime vector to each event, while a covector field assigns a dual vector. Their components at the same event satisfy

$$A'^\mu(x') = \Lambda^\mu{}_\nu A^\nu(x), \quad W'_\mu(x') = (\Lambda^{-1})^\nu{}_\mu W_\nu(x).$$

Here A and W can be independent fields. If $W_\mu = g_{\mu\nu} A^\nu$, the covector field is the metric dual of A .

Theorem 20.3: General tensor field transformation

For a tensor field of type (r, s) ,

$$T'^{\mu_1 \cdots \mu_r}{}_{\nu_1 \cdots \nu_s}(x') = \Lambda^{\mu_1}{}_{\alpha_1} \cdots \Lambda^{\mu_r}{}_{\alpha_r} \\ \times (\Lambda^{-1})^{\beta_1}{}_{\nu_1} \cdots (\Lambda^{-1})^{\beta_s}{}_{\nu_s} T^{\alpha_1 \cdots \alpha_r}{}_{\beta_1 \cdots \beta_s}(x).$$

Upper and lower indices

With the convention

$$V'^{\mu} = \Lambda^{\mu}_{\nu} V^{\nu},$$

every upper index transforms with a factor of Λ . A lower index transforms with the inverse matrix,

$$\omega'_{\mu} = (\Lambda^{-1})^{\nu}_{\mu} \omega_{\nu}.$$

This opposite transformation ensures that the contraction of a covector and a vector remains unchanged,

$$\omega'_{\mu} V'^{\mu} = \omega_{\mu} V^{\mu}.$$

Each index of a tensor transforms separately. For example,

$$T'^{\mu}_{\nu} = \Lambda^{\mu}_{\alpha} (\Lambda^{-1})^{\beta}_{\nu} T^{\alpha}_{\beta}.$$

The notation $\Lambda_{\mu}^{\nu} = (\Lambda^{-1})^{\nu}_{\mu}$ is defined by raising and lowering the matrix indices with the metric. If both input and output covector components are arranged as columns, the corresponding matrix is $(\Lambda^{-1})^{\top}$, not Λ . Index positions and index order must therefore be respected.

Student work – derivation

Derive Theorem 20.3.

20.2 Passive and active viewpoints

The equations above use the passive viewpoint. The geometric event is fixed, while its coordinates and the components used to describe field values change. Write the coordinate change as $x' = F(x) = \Lambda x + a$. Solving for the old coordinates gives $x = F^{-1}(x')$. Thus the scalar-field law can also be written as an equality of functions,

$$\phi'(z) = \phi(F^{-1}(z)) = \phi(\Lambda^{-1}(z - a)),$$

where z is an arbitrary numerical coordinate tuple. The inverse appears because the value expressed in new coordinates must be found at the matching old coordinates. This is still the same field, expressed in two charts.

For an *active* transformation, keep one coordinate chart fixed and move the field configuration by the spacetime map represented by F . Denote the resulting scalar field by ϕ_F . A value originally at x is carried to $F(x)$, so

$$\phi_F(F(x)) = \phi(x), \quad \phi_F(x) = \phi(F^{-1}(x)).$$

The algebraic form is the same, but the interpretation differs. A passive transformation relabels a fixed configuration. An active transformation changes the configuration in a fixed chart. For example, an active spatial translation by a sends $\phi(x)$ to $\phi(x - a)$, moving its profile by $+a$.

For a vector field the active transformation also changes the components,

$$A_F^\mu(x) = \Lambda^\mu{}_\nu A^\nu(F^{-1}(x)).$$

The inverse in the argument and the factor multiplying the field value play different roles. Whether such a transformation maps solutions of a field equation to solutions is a question about that equation, not just its notation.

Student work – derivation

Derive the same-coordinate transformation law for a scalar field.